



DIGITAL VLSI DESIGN

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PART 1

MOS Inverters (Static Characteristics)

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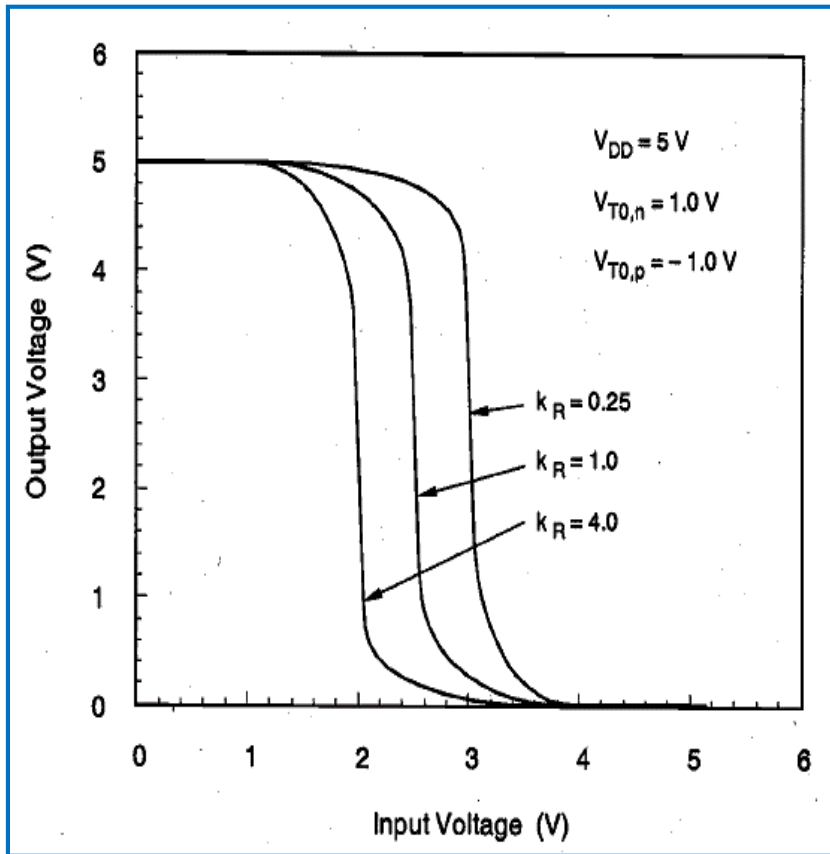
Part e - CMOS Inverter

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MOS Inverters (Static Characteristics)

Effect K_R on VTC



$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{KR}} (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{KR}}\right)}$$

When $V_{DD} = 3V$, $V_{T0,n} = 0.7$ & $V_{T0,p} = -0.7$

$$\text{i) } KR = 1; V_{th} = \frac{0.7 + \sqrt{\frac{1}{1}}(3 - 0.7)}{\left(1 + \sqrt{\frac{1}{1}}\right)} = \frac{V_{DD}}{2} = 1.5$$

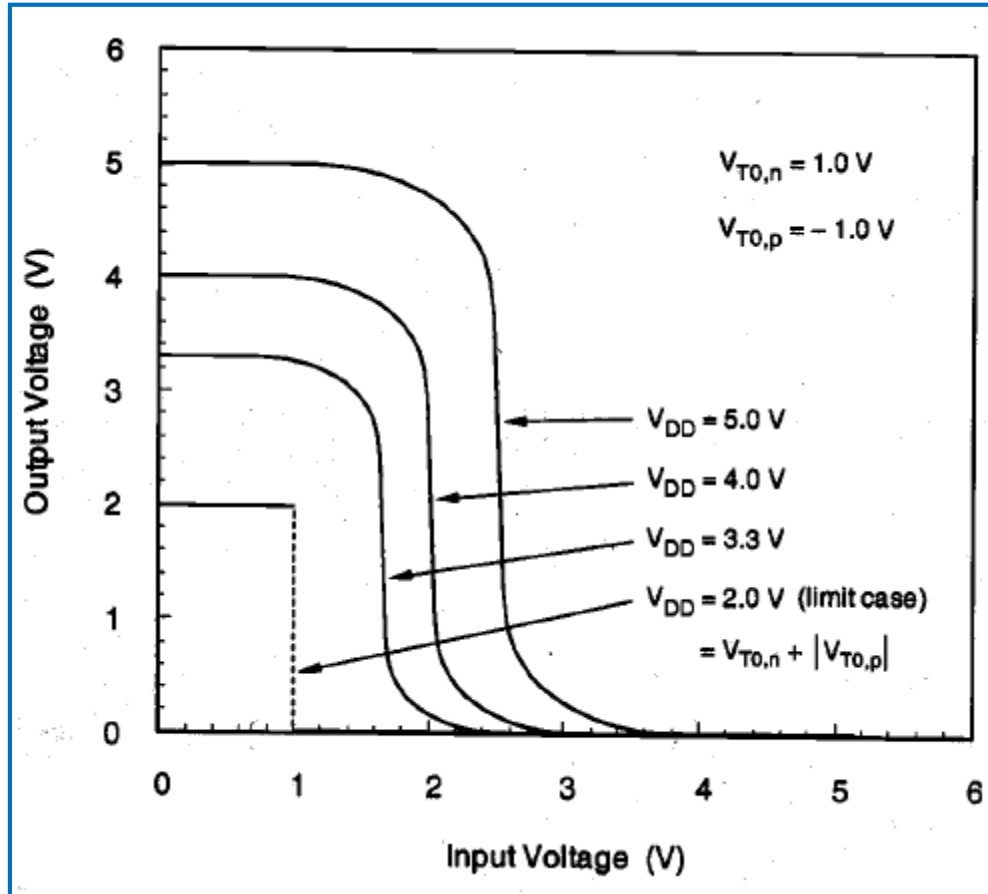
$$\text{ii) } KR = 4; V_{th} = \frac{0.7 + \sqrt{\frac{1}{4}}(3 - 0.7)}{\left(1 + \sqrt{\frac{1}{4}}\right)} = 1.24V$$

$$\text{iii) } KR = 0.25; V_{th} = \frac{0.7 + \sqrt{\frac{1}{0.25}}(3 - 0.7)}{\left(1 + \sqrt{\frac{1}{0.25}}\right)} = 1.76V$$

From this it clear that INCREASE in K_R greater than UNITY will decrease the V_{th} and is always Less than $\frac{V_{DD}}{2}$ & DECREASE in K_R less than UNITY will increse the V_{th} and is always greater than $V_{DD}/2$

MOS Inverters (Static Characteristics)

Effect of Supply Voltage Scaling on VTC



*Power dissipation is strong function of V_{DD} .
As a result scaling Power Supply is one of the important criteria in designing.*

CMOS Inverter continues to operate correctly (by neglecting the sub threshold condition) with scaled supply voltage.

The Supply voltage can be scaled down up to minimum value of

$$V_{DD(\min)} = V_{T0,n} + |V_{T0,p}|$$

Correct inverter operation will sustained if at least one of MOS remains conducting for any given input voltage.

MOS Inverters (Static Characteristics)

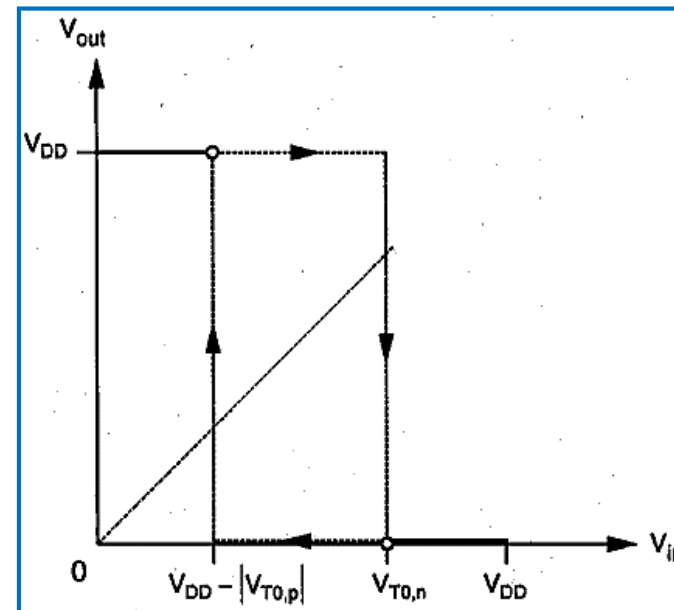
Effect of Supply Voltage Scaling on VTC

If Supply voltage is made
 $V_{DD} < V_{T0,n} + |V_{T0,p}|$

Then, CMOS inverter VTC behaves somewhat differently as it will contain a region in which NONE of the transistors are conducting

The output voltage within this region is then determined by previous state of the output since previous state is always preserved as stored charges at the output node.

The VTC exhibits hysteresis behavior for very Low Supply voltage



MOS Inverters (Static Characteristics)

Example 5.4

Consider a CMOS inverter circuit with the following parameters :

$$V_{DD} = 3.3 \text{ V}$$

$$V_{T0,n} = 0.6 \text{ V}$$

$$V_{T0,p} = -0.7 \text{ V}$$

$$k_n = 200 \mu\text{A/V}^2$$

$$k_p = 80 \mu\text{A/V}^2$$

Calculate the noise margins of the circuit. Notice that the CMOS inverter being considered here has $k_R = 2.5$ and $V_{T0,n} \neq |V_{T0,p}|$; hence, it is not a symmetric inverter.

MOS Inverters (Static Characteristics)

First, the output low voltage V_{OL} and the output high voltage V_{OH} are found, using (5.54) and (5.55), as $V_{OL} = 0$ and $V_{OH} = 5$ V. To calculate V_{IL} in terms of the output voltage, we use (5.62).

$$V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$$

$$= \frac{2V_{out} - 0.7 - 3.3 + 1.5}{1 + 2.5} = 0.57V_{out} - 0.71$$

Now substitute this expression into the KCL equation (5.59).

$$\frac{k_n}{2} \cdot (V_{in} - V_{T0,n})^2 = \frac{k_p}{2} \cdot \left[2 \cdot (V_{in} - V_{DD} - V_{T0,p}) \cdot (V_{out} - V_{DD}) - (V_{out} - V_{DD})^2 \right] \quad (5.59)$$

MOS Inverters (Static Characteristics)

$$2.5(0.57 V_{out} - 0.71 - 0.6)^2 = 2(0.57 V_{out} - 0.71 - 3.3 + 0.7)(V_{out} - 3.3) - (V_{out} - 3.3)^2$$

This expression yields a second-order polynomial in V_{out} , as follows:

$$0.66 V_{out}^2 + 0.05 V_{out} - 6.65 = 0$$

Only one root of this quadratic equation corresponds to a physically correct solution for V_{out} (i.e., $V_{out} > 0$).

$$V_{out} = 3.14 \text{ V}$$

From this value, we can calculate the critical voltage V_{IL} as:

$$V_{IL} = 0.57 \cdot 3.14 - 0.71 = \underline{\underline{1.08 \text{ V}}}$$

To calculate V_{IH} in terms of the output voltage, use (5.67):

$$\begin{aligned} V_{IH} &= \frac{V_{DD} + V_{T0,p} + k_R \cdot (2 V_{out} + V_{T0,n})}{1 + k_R} \\ &= \frac{3.3 - 0.7 + 2.5(2 V_{out} + 0.6)}{1 + 2.5} = 1.43 V_{out} + 1.17 \end{aligned}$$

Next, substitute this expression into the KCL equation (5.64) to obtain a second-order polynomial in V_{out} .

$$2.5[2(1.43 V_{out} + 1.17 - 0.6) V_{out} - V_{out}^2] = (1.43 V_{out} - 1.43)^2$$

$$2.61 V_{out}^2 + 6.94 V_{out} - 2.04 = 0$$

Again, only one root of this quadratic equation corresponds to the physically correct solution for V_{out} at this operating point, i.e., when $V_{in} = V_{IH}$.

$$V_{out} = 0.27 \text{ V}$$

From this value, we can calculate the critical voltage V_{IH} as:

$$V_{IH} = 1.43 \cdot 0.37 + 1.17 = \underline{\underline{1.55 \text{ V}}}$$

MOS Inverters (Static Characteristics)

Finally, we find the noise margins for low voltage levels and for high voltage levels using (5.3) and (5.4).

$$NM_L = V_{IL} - V_{OL} = 1.08 \text{ V}$$

$$NM_H = V_{OH} - V_{IH} = 1.75 \text{ V}$$

MOS Inverters (Static Characteristics)

5.6 Consider a CMOS inverter with the following parameters:

<i>nMOS</i>	$V_{T0,n} = 0.6 \text{ V}$	$\mu_n C_{ox} = 60 \mu\text{A/V}^2$	$(W/L)_n = 8$
<i>pMOS</i>	$V_{T0,p} = -0.7 \text{ V}$	$\mu_p C_{ox} = 25 \mu\text{A/V}^2$	$(W/L)_p = 12$

Calculate the noise margins and the switching threshold (V_{th}) of this circuit. The power supply voltage is $V_{DD} = 3.3 \text{ V}$.



THANK YOU

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